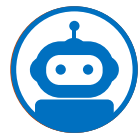
Human Imitator

Hi, I manage an online electronics store selling phone chargers. We have unpredictable daily demand that follows a Poisson distribution with λ=6 units/day. If we don't meet demand, customers will leave without purchasing anything. Holding costs us \$0.50/unit/day, while each unfulfilled order incurs a penalty of \$2.75/unit/day. There's also a fixed cost of \$3.00 per shipment and a 4-day delivery delay. Our storage capacity allows up to 80 units at any given time, and orders can be placed in increments of no more than 30 units. Could you assist me in developing an optimal inventory strategy considering these parameters?

What planning time horizon (in days) should we optimize the inventory strategy for? For example, 30 days, 90 days, or 365 days.



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Let's assume a 90-day planning period. This aligns well with our seasonal sales patterns and accounts for potential fluctuations over the quarter.

Last question, about risk preference, on a scale from -10 (very risk-averse) to 10 (very risk-seeking), how would you rate your business strategy?

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On this scale, my approach leans towards moderate risk aversion. A score of 3 reflects a cautious stance where stability and reliability are prioritized over aggressive growth strategies. This rating ensures that we maintain sufficient stock levels to satisfy customer needs while minimizing unnecessary inventory buildup.